

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1403

Loon Lake 100 kV
March 16, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1403. The Interconnection Customer proposes a 104.2 MW solar photovoltaic facility interconnecting at the Loon Lake 100 kV substation in the Illinois area. The request was received on February 10, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 508 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$149,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Houma-French Landing Dam Substation 100kV L	Mount Polley-Mosby 100kV	108.7	122	9.7
Mount Polley-Pampanos 100kV	Tidelans 230/1kV TX	106.6	149	13.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Houma-French Landing Dam Substation 100kV L	Reconductor line	\$437,000	16% (\$70,000)
Mount Polley-Pampanos 100kV	Reconductor line	\$422,000	19% (\$79,000)
	Total allocated to project		\$149,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.